

# What makes a sketch-based long-read mapper cheaper: nine changes to shmap, by the layer they act on

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## Abstract

**Summary:** shmap-rs is a Rust port of *map-shmap* [1] that is 2.14–2.97× faster single-threaded and uses 7.0–7.3× less peak memory while producing byte-identical output. This companion note accounts for where that came from. 9 deliberate changes are responsible, together with the rewrite that carries them — and the ablation below shows the rewrite is the larger of the two. All 9 of the changes preserve output exactly, none alters a mapping decision, and each is compared against the upstream source it replaces at a pinned revision. They are presented in the order they had to be discovered: the cost structure the algorithm implies, the data structure that structure argues for, the algorithmic identities that let work be skipped, the parallel decomposition, and only last the code. That order is the result rather than the presentation: the largest wins are in the first two layers, and instruction-level work — the layer that usually attracts attention — is worth least here, and is where four further attempts measured negative.

**Availability:** original <https://github.com/pesho-ivanov/shmap>; this port <https://github.com/d0bromir/shmap-rs>, MPL-2.0, version 1.4.1.

## 1 The theoretical base

*map-shmap* [1] is Ivanov and Medvedev’s, and nothing below changes what it computes. It sketches read and reference by FracMinHash, attributes each seed match to the overlapping fixed-width buckets whose window contains it, prunes buckets with an admissible seed heuristic, and refines the survivors with a sliding window. Its per-read cost has three factors — reads, matches examined, and a logarithmic term in reference size — and the important property for an implementer is which of them is paid in *arithmetic* and which in *memory traffic*.

Almost all of it is memory traffic: seeds are looked up in an index far larger than the last-level cache, the buckets they touch are scattered across an array sized by the reference, and the refinement sweep walks a window last touched by a different read. Profiling agrees — **map: match\_rest** is the costliest stage, 21.0% of CPU on average, and pointer-chasing rather than arithmetic.

Two consequences order everything else. A change that reduces *how much memory is touched* outranks one that reduces instructions issued, because the machine is waiting either way; and the sizes that matter are those set by the *reference*, since a whole-genome run cannot escape them — which points at the accumulator.

## 2 The data structure

The C++ allocates one dense bucket array per reference segment, sized so that every window the algorithm

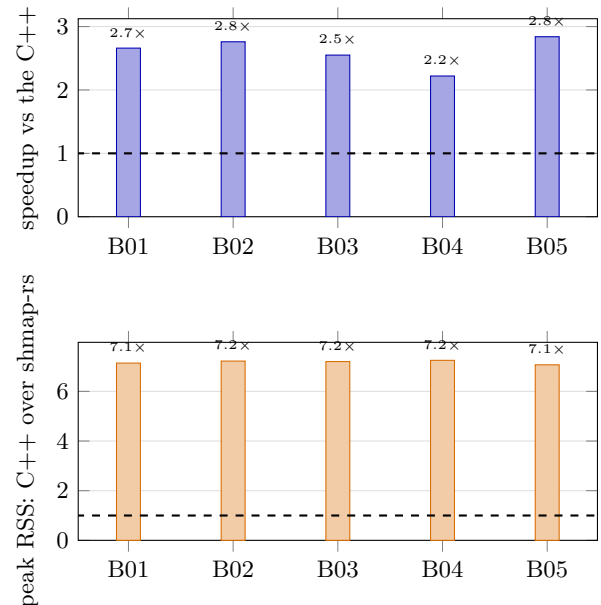


Figure 1: The two headline results, per benchmark, single-threaded on both sides: wall-clock speedup over the C++ (top) and how many times less peak memory the port uses (bottom). The dashed line is parity.

could ever use has a slot (**buckets.h**, Table 1). The width is a compile-time minimum rather than the width any particular read will use, so a human genome costs about fifteen gigabytes before a single read is mapped, and that cost is paid whether one read is mapped or millions.

The observation that removes it is that *a read can only touch buckets within its own span*. The array does not have to be indexed by reference bucket; it has to be indexed by the buckets this read can reach, which is set by the read’s own half-length. shmap-rs sizes it that way and keeps a sparse map as the fallback, and the two paths are tested to agree exactly. The footprint stops being a property of the genome and becomes a property of the query: 2.58–2.67 GB single-threaded against 18.85 GB — the lower panel of Figure 1, and the reason the C++ figure is flat there is that it never depended on the work being done.

One change, in one data structure, and it is most of the memory result. It also removed a contention bottleneck that had made multithreading nearly useless: a per-worker copy of a genome-sized array does not scale, and no thread tuning would have rescued it. The data structure had to be right before the parallel decomposition could be.

| #                     | Change                                                                                   | C++ replaced                   | Effect, as measured when it landed                                                                                                                                                                                                                                                           | Exact |
|-----------------------|------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <i>data structure</i> |                                                                                          |                                |                                                                                                                                                                                                                                                                                              |       |
| 1                     | Dense bucket array sized by the read's own half-length, not the reference                | <code>buckets.h:76-97</code>   | ~4 MB/worker vs the C++'s constant ~15 GB — the 6.9-7.4x-less-memory headline (RESULTS.md §1) is almost entirely this. On a 6,000-read whole-genome run: <code>bucket_merge</code> 1342.1 s → 39.1 s, whole run 1995.6 s → 725.6 s                                                           | yes   |
| <i>algorithm</i>      |                                                                                          |                                |                                                                                                                                                                                                                                                                                              |       |
| 2                     | Streaming multi-hit seeds                                                                | <code>shmap.h:105-140</code>   | Not isolated as its own standalone percentage in the record; replaces ~O(hits) hashmap inserts with O(1) integer work per hit (§2)                                                                                                                                                           | yes   |
| 3                     | <code>RefineCache</code>                                                                 | <code>shmap.h:508-520</code>   | 44% of <code>find_best_mapping</code> calls eliminated, flat across 1x/3x/10x coverage: <code>match_rest_for_best2</code> <b>-66 to -67%</b> , <code>refine</code> <b>-39 to -41%</b> , <code>query_mapping</code> <b>-9 to -11%</b> (§3)                                                    | yes   |
| 9                     | Final bucket ordering: an unstable sort on a packed key reproduces stable-sort output    | <code>buckets.h:151-174</code> | Sorts 8-byte packed keys instead of 32-byte records — no isolated standalone percentage in the record, but real: moving 4x fewer bytes and skipping a stable sort's temporary allocation                                                                                                     | yes   |
| <i>parallelism</i>    |                                                                                          |                                |                                                                                                                                                                                                                                                                                              |       |
| 4                     | Multithreaded read mapping                                                               | — <i>none</i>                  | RESULTS.md §2 (single-thread and <b>-04</b> per benchmark) and §3 (the full thread-scaling table, including the NUMA ceiling Q4/Q5 diagnosed)                                                                                                                                                | yes   |
| 5                     | Hash-sharded index storage + chunked reference sketching, both parallel across <b>-0</b> | — <i>none</i>                  | <code>index_initializing</code> <b>8.4 s → 1.1-1.8 s</b> ; chunked sketching <b>~18% off 'indexing'</b> at <b>-08</b> (§5)                                                                                                                                                                   | yes   |
| 6                     | Two-pass parallel FASTA parsing                                                          | — <i>none</i>                  | <code>index_reading</code> <b>4.4 s → 1.5-1.7 s</b> (§6)                                                                                                                                                                                                                                     | yes   |
| <i>code</i>           |                                                                                          |                                |                                                                                                                                                                                                                                                                                              |       |
| 7                     | Sketching/index hot loop                                                                 | <code>sketch.h:461-463</code>  | <b>~13-17% off 'index_sketching'</b> is the bounds-check-free iteration alone (§7); the other three items in this row have no isolated standalone percentage in the record                                                                                                                   | yes   |
| 8                     | Lower allocation/memory traffic                                                          | <code>types.h:43-71</code>     | <b>~11-13% off peak RSS</b> is the bounded read-ahead alone; <b>~5% wall</b> is <code>lto="fat"</code> alone (§8); the rest are allocation- <i>count</i> reductions (e.g. ~300 M fewer heap allocations for single-occurrence seeds) without an isolated standalone percentage in the record | yes   |

Table 1: Every change that makes shmap-rs cheaper than the C++, grouped by the layer it acts on, with the upstream source each replaces at commit 63f1103. Effects are what the change was worth against the build it landed on and are deliberately not restated against the current one. They do not sum: the whole-run comparison is measured separately. *Exact* means the mapping output is byte-identical.

### 3 The algorithmic identities

Three changes skip work by proving it need not be done. None is an approximation; each is an identity, and the byte-identical output is the check.

**The second-best search repeats the first.** Computing mapq needs the second-best mapping, and the C++ searches for it with a second full pass over the same surviving buckets (`shmap.h`, Table 1). On the two fixed-length metrics, scoring a bucket turns out to be a pure function of its *location*: it never reads the mutable state that pruning updates between the passes, and it restores its own scratch exactly. The second pass can therefore replay the first's scores. It is deliberately not applied to the two bucket-based metrics, where the score is built out of exactly that mutable state, so the identity does not hold. Because the acceptance threshold ratchets upward during the first pass, a bucket discarded late there can still clear the second pass's flatter cutoff — so the win is partial by construction, which is why it is roughly two calls in five and not all of them.

**A repetitive seed does not need a hash map.** The C++ builds a fresh scratch hash map per seed to accumulate its hits; the hits are already sorted, which makes the map redundant, and the same buckets can be accumulated in place at constant integer cost per hit.

**An unstable sort can reproduce a stable one.** Ordering the final buckets by descending match count is the original paper's own Algorithm 4 "Optimization 2". Packing the tie-break into the sort key makes an unstable sort reproduce exactly what a stable sort would give, over keys a quarter the size of the records. Here

the port is stronger than the source rather than merely faster: the C++'s `std::sort` guarantees no particular order among equal keys, even between its own runs, and this does.

### 4 The parallel decomposition

3 of the 9 exist only because the C++ is single-threaded, so they replace nothing upstream: threaded read mapping, hash-sharded index storage with chunked reference sketching, and a two-pass count-then-fill FASTA reader. The constraint that shapes all three is that output must not depend on thread count, which rules out any decomposition whose result depends on completion order. A sketch window is a pure function of the bases under it; a shard's contents are determined by the hash and not by which worker filled it; the FASTA reader's second pass writes exactly what its first counted. Determinism is checked rather than assumed, on every thread count and benchmark (15 of 15 checks).

The ceiling is again memory, not scheduling: on the four-socket machine scaling stops near 6.2×, while a single-NUMA-node machine running the identical binary continues to 12.0×. Per-node index replication was implemented five ways to remove it and net-regressed every time.

### 5 The code

Only two of the 9 are instruction-level, and they are last because they are worth least. The sketching hot loop precomputes its rotation tables, iterates without bounds checks, hashes once per lookup instead of twice,

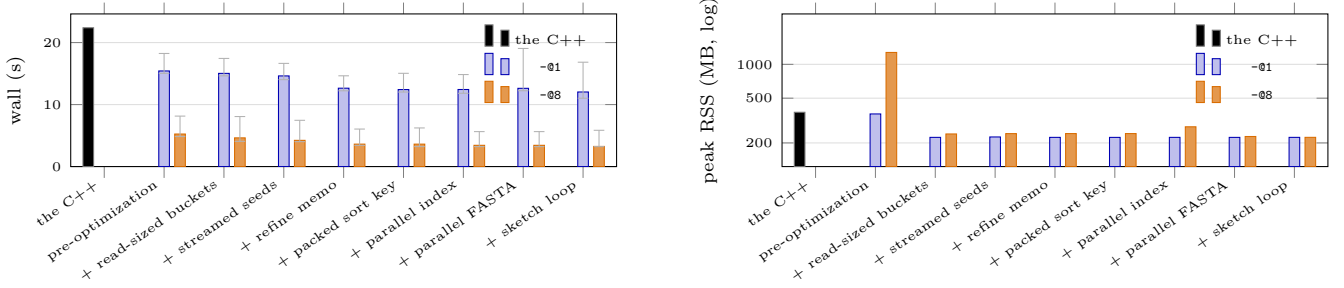


Figure 2: Every optimization, put back one at a time. The leftmost bar is the C++ itself; *pre-optimization* is this port with every ablatale change switched off (SHMAP\_ABLATE), so the step between those two is the port and nothing else — the largest step in the figure, and the one the layered story does not cover. Each rung to the right switches exactly one more change back on, so adjacent rungs differ by one change and nothing else. The two series are the two thread counts, and the gap between them is row 4; row 8 is not ablatale, and the C++ is single-threaded so it has no  $-0N$  counterpart. Left: wall clock, median over 9 round-robin runs of the whole ladder, whiskers at min-max, so an unresolved step is visible as one. Right: peak resident set, log scale — one worker’s genome-sized accumulator hides under the index-build peak,  $-08$  workers’ copies do not. Every rung’s mapping is byte-identical; the run fails otherwise. The C++ bar carries no  $-x$  (the C++ has no equivalent) while the rungs do, costing  $1.06\times$ , so the first step is understated rather than flattered. Numbers and per-stage timers in `figure_ablation.tsv`; the decomposition against the C++ in `reconcile.tsv`.

and sizes its output buffer from the actual distribution. The allocation work stores a single query position inline rather than in a one-element vector, has matches borrow their seeds instead of copying them, indexes the difference histogram densely, and discards the reference sequence after sketching, where the C++ keeps a second full copy of the genome for the whole run.

Their effects are in Table 1: real, but small beside the first two layers. The stronger evidence for the ordering is negative — hand-written SIMD sketching, two-bit packing, software prefetching of the pruning lookups and per-node index replication were each built or probed at full scale and each measured a loss or a wash. The sketching loop is load-port bound at six level-one loads per base, so more lanes add pressure where there is none to spare, and packing removes the wrong two of those loads. When a workload waits on memory, instruction-level effort is the layer with least left in it.

## 6 What each one is worth

Table 1’s effects were each measured against the build that change landed on, months apart and on different inputs. They share no baseline, machine or read set and do not sum, so the table cannot say how much of the result is which change. Figure 2 answers that.

The result being decomposed is Figure 1: at one worker, where the comparison is like for like, the port is  $2.14\text{--}2.97\times$  the C++ on the whole genome and uses  $7.0\text{--}7.3\times$  less peak memory. The ladder below is measured on a chromosome, for the reason its own paragraph gives, so its absolute ratios are smaller; what it contributes is which change earned which part.

Every change is switchable at run time, in one instrumented build, back to the code path it replaced. The leftmost bar has all 7 of them off; each bar to the right turns exactly one more back on, in the layer order argued above; the rightmost is the shipped build. Adjacent bars therefore differ by one change and noth-

ing else — one binary, one machine, one compiler, one input — which is what makes a step attributable, and attributable to that change *given the ones to its left*: what the parallel index is worth depends on whether each worker still carries a genome-sized accumulator. Every bar’s mapping is compared byte for byte against the leftmost — a stricter test of the exactness claim than the suite’s, because it holds the input fixed and varies only the optimization. Those switches are not in the released mapper: a branch and a scratch buffer that exist only to be measured are paid for by everyone who runs the program, so they live on an archived, never-merged branch, and the harness refuses a build that lacks them rather than measuring one binary 7 times and drawing a flat ladder.

2 of the 9 cannot be a rung and are reported as absent rather than estimated: threaded read mapping is a capability rather than a branch, so it is the gap between the two series, and the allocation-and-layout row is type-level, as below.

End to end the ladder is  $1.28\times$  at one worker and  $1.62\times$  at 8, for  $1.62\times$  and  $5.69\times$  less peak memory; the largest single step is refine memo (row 3), worth 14.4% of the bar before it. All of it is chromosome scale on LAPTOP-6S4BOT55, a laptop with 8 cores, because the pre-optimization accumulator does not fit there once, let alone per worker.

**The first bar is the largest, and it is not one of the nine.** The step from the C++ to *pre-optimization* — the port itself, before any ablatale change is switched back on — is  $1.67\times$ , against  $1.36\times$  for the whole ladder to its right; the two compose by multiplication and return the  $2.27\times$  measured directly between the outer bars, which is what makes the decomposition checkable. Two things sit in it. One is the allocation-and-layout row: a numbered change, but type-level — matches borrow their seeds, the single-occurrence case is stored inline — so reversing it is a different program, not a different branch, which is why

it cannot be a rung. The other is the rewrite itself: a different language, allocator and standard library. This note does not separate them, and doing so needs type surgery rather than a switch. It does rule out the cheap explanation: the same source at the stock release profile, without fat link-time optimization, measures  $0.98\times$  the shipped build — indistinguishable from it — so the bar is not the Rust being built harder.

**The memory panel shows the mechanism, not the magnitude.** It has to: what the accumulator change removes is a slot per 5 bases *of the reference*, about a fifth of a gigabyte on the chromosome this figure uses and over ten on the genome. So the panel reads  $1.62\times$  at one worker where the whole-genome comparison above reads  $7.0\text{--}7.3\times$ , 18.85 GB against 2.58–2.67 GB. They are not in tension: the smaller is the same effect measured where it is small. The panel shows its *shape*, which is the argument — the ratio grows with worker count, to  $5.69\times$ , because the array removed was per worker. The genome-scale evidence needs no ablation and is already in the suite: the C++’s peak RSS moves by 6 MB across every benchmark, whose read sets differ by more than an order of magnitude in size. A footprint that does not notice its input belongs to the reference.

## 7 What none of them changed

All 9 preserve output exactly, enforced rather than claimed: the mapping is byte-identical across every thread count and against the C++, ground-truth accuracy is unchanged at 98.76–99.21%, and any drop in mapped or confident reads blocks a merge. Table 1 is generated from the repository’s own account of what changed, each row citing upstream at [63f1103](#).

## References

- [1] Ivanov,P. and Medvedev,P. map-shmap: practical long-read mapping with a seed heuristic on sketches. Manuscript in preparation.